

한국건강증진개발원, 대한예방의학회 공동개최

2023 스마트 건강관리 서비스 세미나

일 시: 2023.11.23. (목) 10:30~17:00

장 소: 인천 송도 컨벤시아 회의실6

주제

디지털 시대의 지역사회 건강증진 재조명

주요내용

- 스마트 건강관리 서비스 연구개발 현황과 서비스 확장 가능성 및 정책반영에 대한 현실적 문제점 발제와 토론
- 우리나라 디지털 헬스케어 국가사업 추진 현황 및 비전 공유



2. 디지털 기술을 활용한 플랫폼 노동자의 스마트 건강관리

연세대학교 윤진하 교수



Gig Economy Workers and Platform Labor

... the business model has common elements

The diagram illustrates the business model for gig economy workers. It shows a central 'Business model' box connected to several other boxes: 'Indicators for matching clients and workers', 'Allocation of work', 'Determining pay rates', 'Recruitment and matching', 'Revenue model', 'Monitoring, tracking and evaluating workers through digital tools and algorithms', 'Work process and performance management', and 'Rules of governance'. The 'Indicators for matching clients and workers' box lists: Client review (60%), Ratings (50%), Worker profile (46%), Project history or portfolio (27%), and Rate assessed by the worker (21%). The 'Allocation of work' box lists: Unilaterally determined by platforms, Exclusionary clauses, Deactivation of accounts, and Dispute resolution. The 'Revenue model' box lists: Commission fees and subscription plans are integral to platform revenue model. The 'Monitoring, tracking and evaluating workers through digital tools and algorithms' box lists: Ratings, Refusing work, and Monitoring, tracking and evaluating workers through digital tools and algorithms. The 'Work process and performance management' box lists: Work process and performance management. The 'Rules of governance' box lists: Rules of governance. The diagram is credited to 'ILO.org'.

“ (ILO 2021 report) ”

Digital labour platforms are growing and presenting both opportunities and challenges to workers.

Common element for matching clients and workers

- Allocation of worker
- Determining pay rates

Unilateral determination of social agreement by platforms

Monitoring, tracking, and evaluation of workers through digital tools and algorithms

5x rise since 2010

The bar chart shows the concentration of platforms in different countries. The y-axis represents the percentage of platforms, and the x-axis represents the countries. The data is as follows:

Country	Percentage
USA	29%
India	8%
United Kingdom	5%

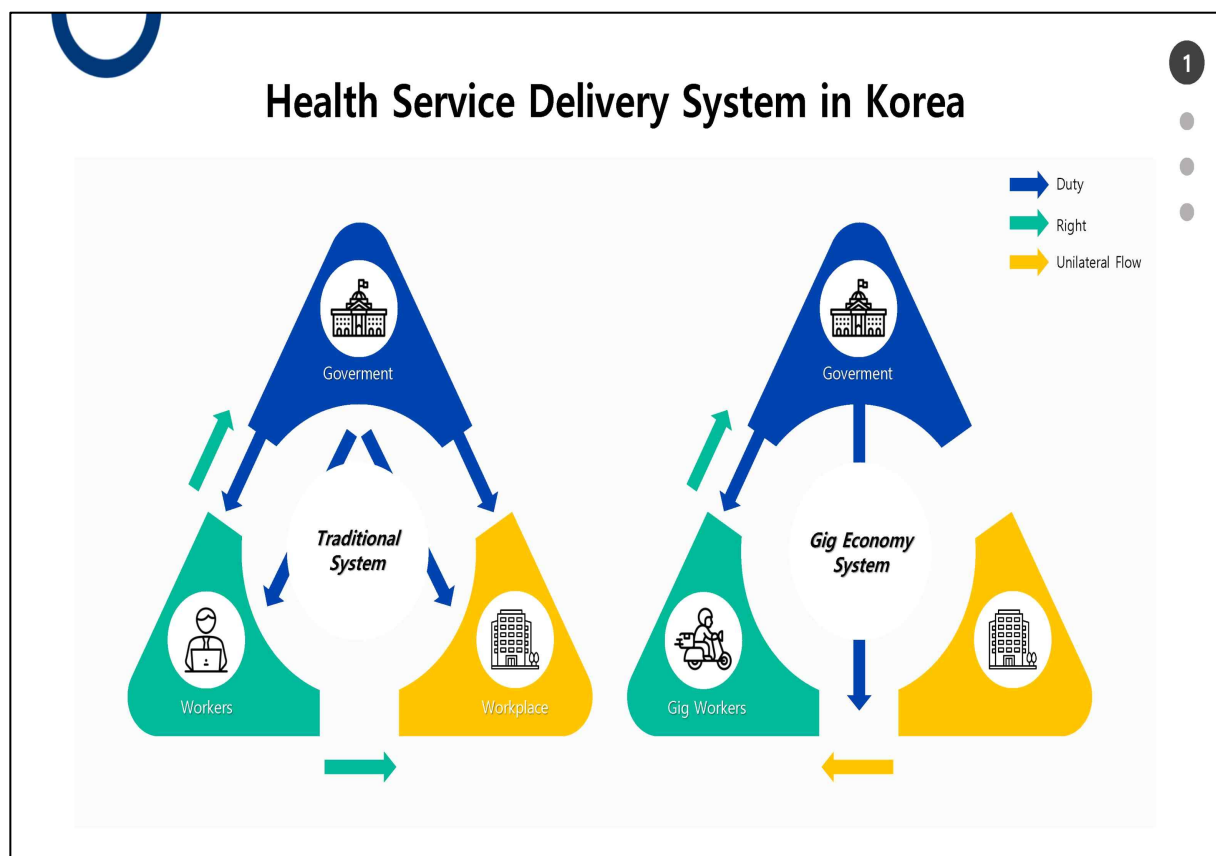
Legend: Hybrid (blue), Delivery (green), Taxi (red), Online web-based (yellow). The chart shows exponential growth in the number of platforms from 2000 to 2020.

같은 공간, 같은 시간, 같은 생각



같은 공간,
같은 시간,
같은 생각

Summary and Status Review		
	Details	Note
Research on the Actual condition	1. Proposal for 7 major platforms for industrial health service and education policy (in downtown Seoul) 2. A fact-finding survey to prepare measures to protect platform workers (Iyeon Jang, Ministry of Employment and Labor) 3. Occupational safety and health fact-finding survey, development of survey tools (Seung-Hyun Lee, KOSHA)	- Conduct continuous monitoring and discuss process to build data - Survey of Platform worker Occupational Health service Needs and Priorities
Harmful environment assessment	1. Labor environment Hazardous and dangerous environment, job stress, safety (Seung-Hyun Lee, KOSHA) 2. Long working hour 3. Study on Platform Working Hours for Courier Drivers (Tae-Won Jang, Ministry of Employment and Labor) 4. Study on Platform Working conditions and research on ways to prevent health problems such as death from overwork (Hee-Jung Lee, KOSHA)	- Hazard risk factors by job category and Hazardous substance exposure level assessment - Development of standard working model of a platform labor (focused on the Musculoskeletal disorders) - Development of platform workers job stress assessment tool
Health care	1. Occupational Health Organization 2. Service Delivery Method 3. Service contact problem 4. Funding 5. Medical Check-Health Management 6. Chronic disease Management 7. Long working hour, Overwork	- Selection of medical check-up items - Development of medical check-up system - Health management method and subject development - Construction and presenting utilization model
Occupational Health Service Delivery System	1. Funding Pilot Study (Public Workers Solidarity Foundation) 2. Service delivery system using IT (Korea Occupational Safety and Health Agency, Future Institute) 3. Health management through data sovereignty, my data business (Sanggug Lee, Ministry of Land, Infrastructure and Transport)	- Building occupational health services that can be delivered to platform worker
Legal protection system	1. Enlargement complement 2. Operation of an advisory committee on employment types for worker characteristics (Ministry of Employment and Labor) 3. Industrial accident, employment insurance, vocational training (Ministry of Employment and Labor) 4. A study on the criteria for judgment standard and protection of special type workers (CHUN YUN-KU, Kyonggi Univ.) 5. Platform Worker Legislation Reorganization, Overseas Cases (Eunjeong Park/Kang Jik Song, Korean Society Of Labor Law) 6. Offer through a new system 7. Proof of experience career and support for mutual aid associations (Platform Workers' Mutual Aid Society sailing)	- Sustainable financial preparation

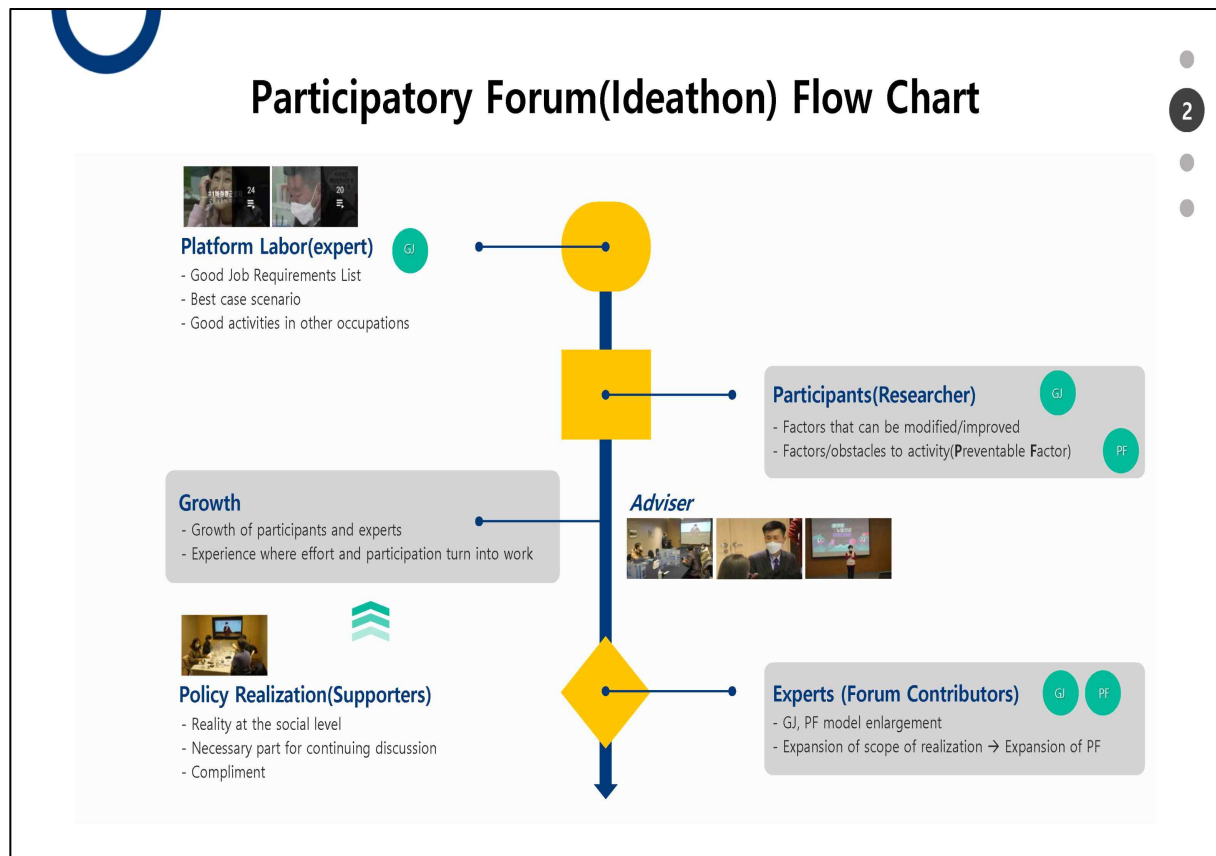




GigUpHs

당신이 원하는 서비스는 무엇인가요?

(인터뷰영상자료)



(회의영상자료)

Participatory Forum(Ideathon) Flow Chart



2

	Status and Problems	Improvements and Necessities
House-keeper	- Subjects excluded from subscribing to industrial accident compensation insurance - Medical check-up: national medical check-up / personal insurance - Difficult to grasp work environment problems. - Eating problems - Musculoskeletal disorders - Emotional labor	- Labor cooperatives - Musculoskeletal disorder management program - Safety education (fall accidents, etc.) - Work standardization - Description for medical check-up - Awareness improvement campaign
Replacement driver	- Quality of sleep - Eating problems - Musculoskeletal disorder - Mental health care	- Musculoskeletal & Mental health management program - Team learning (providing a sense of belonging)
Rider	- Delivery system to encourage speeding and accidents - Customer rating system - Not recognized as an employee - Excessive insurance premium (6 million won/year) - Eating problems & eating habits - Administrative issues in the registration process for industrial accidents - Reasons for difficulty in organizing - New problems emerging in COVID-19	- Mutual Aid Society - No-call campaign to comply with meal-times - Self-examination and screening incentives - Health-related investigation required - Safety rate system (KRW 4,000/1 case)

Participatory Forum(Ideathon)

2

1

How to collect on the Web App

- Platform worker online place (App)
- Worker app/consumer app linked campaign
- Expected effect
 - Integrated management of platform labor force
 - Personal health history (behavior) based service
 - Use for policy improvement

2

How to collect on the Web App(Add On)

- Responding to health issues: on/offline points of contact
 - Shelter for mobile workers
- Responding to safety issues: Existing app add-on method
- Formation of platform council

3

How to use On/Offline Points

- Establish an online space to provide integrated shelter information
- Rest time guarantee campaign
- Medical staff visits counseling center for medical check-ups for platform workers

4

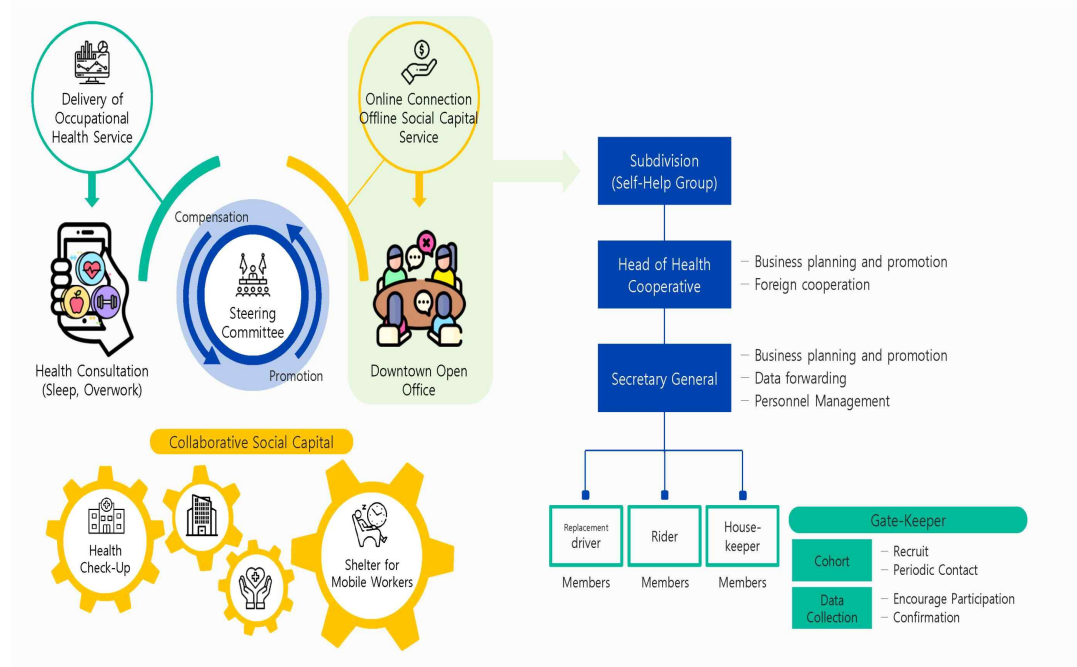
Public Data Utilization Plan

- Safety protection of platform workers through Open-API
- Plans to secure financial resources
- Protection model
 - Online education
- Establishment of information delivery system
- Preventing from overwork
- Offering Incentive



Installing Health System Into Cooperative Self-Help Group

2

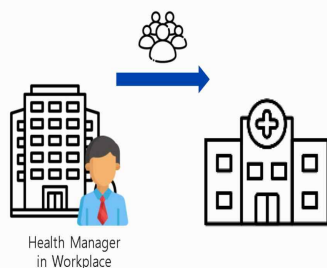


Workers Health Examination(General H.E)

2

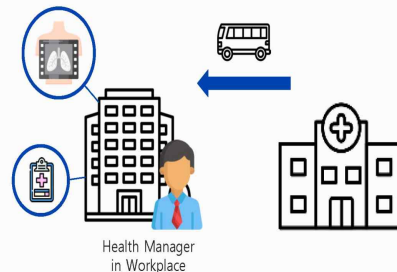
1st Scenario

Worker visits hospital together with or without colleague. The schedule and status of health examination were managed by occupational health manager in workplace.



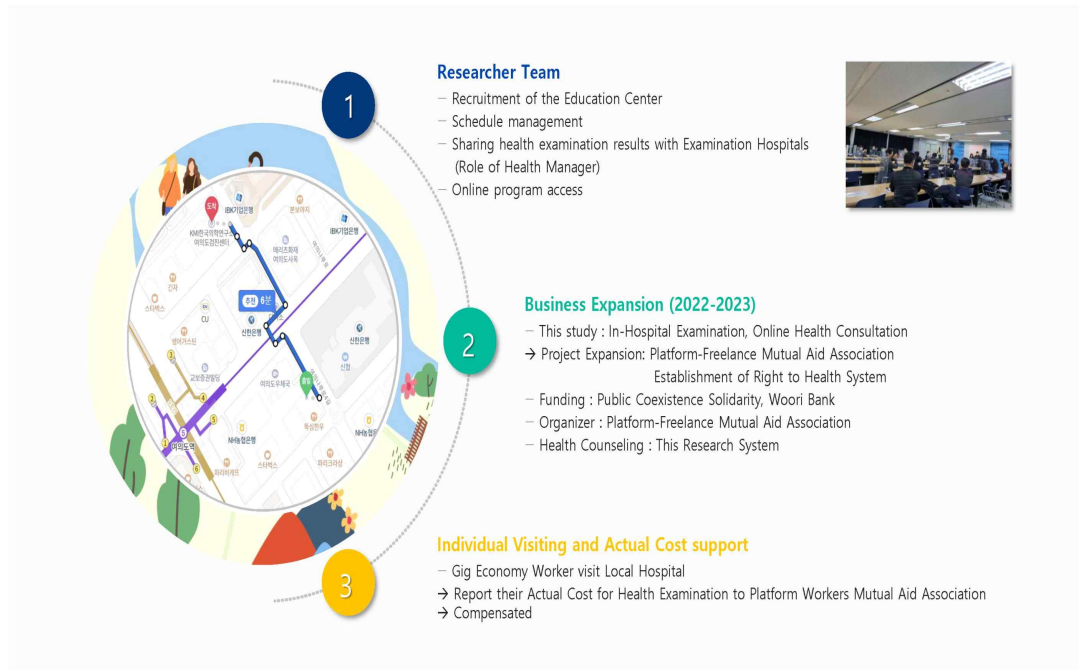
2nd Scenario

Hospital system with Bus visit factory (workplace) and take health examination, and the schedule and status of health examination were managed by occupational health manager in workplace.



Workers Health Examination; 1st Scenario

2



Workers Health Examination; 1st Scenario

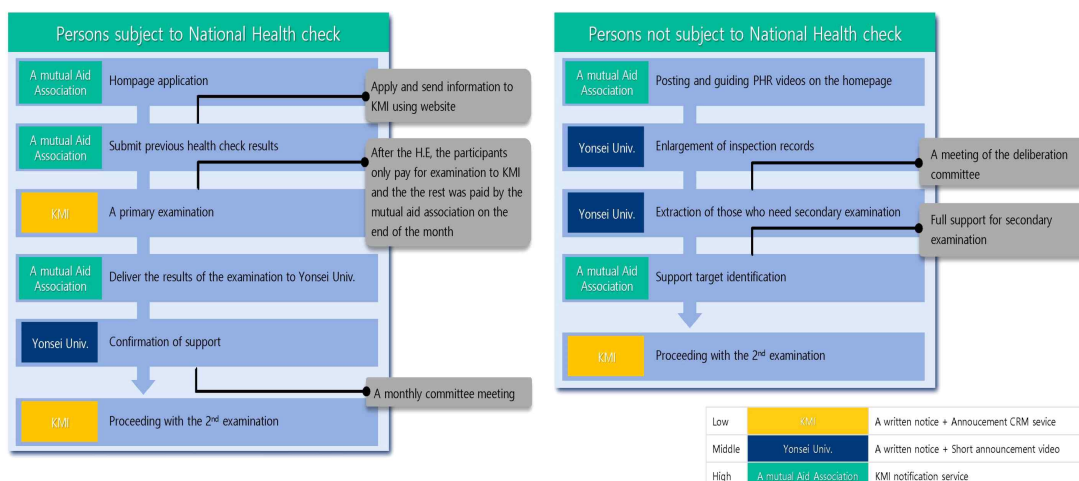
2



2

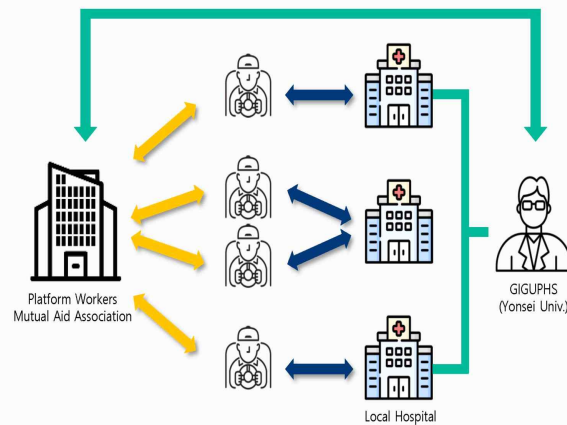
Continuation of Research Achievements into other Projects

- A meeting held with the base hospital, Mutual Aid Association, and GIGUPHS



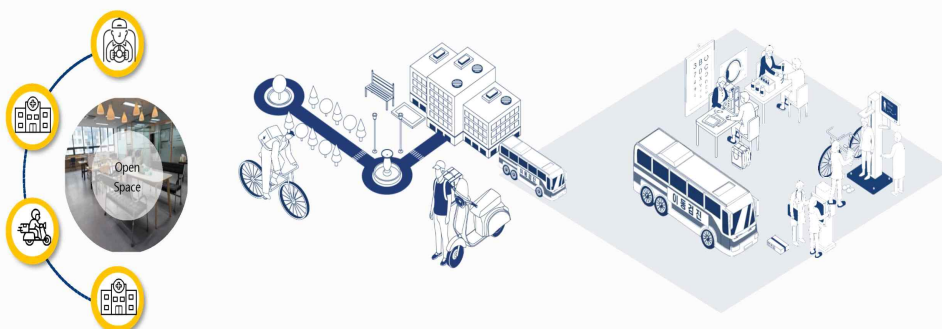
Workers Health Examination; 1st Scenario

2



Workers Health Examination; 2nd Scenario

2



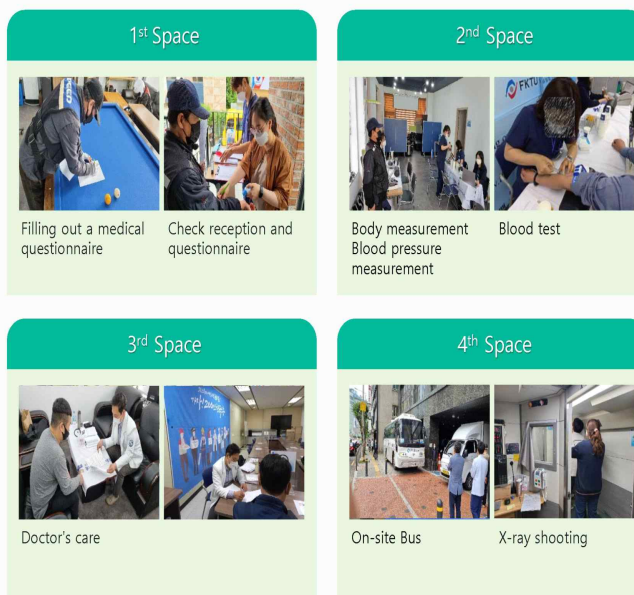
“

Gig Economy Worker get together with other gig economy workers in open space (instead of workplace of standard worker), the hospital system visit with bus.(More than 50 worker is minimal criteria for private hospital system to fulfill the business model) We used lecture room in Resting shelter for outdoor moving workers.

”

We used resting shelter as meeting place among gig economy workers and private hospital for health examination.

2



2

직종별 건강진단

신경계, 심혈관계, 위장관계, 내분비계 및
뇨생식기관별 발병위험도 평가의 건강진단

총 비용 약 73,000원 중
₩ 60,000원 지원
80% 지원

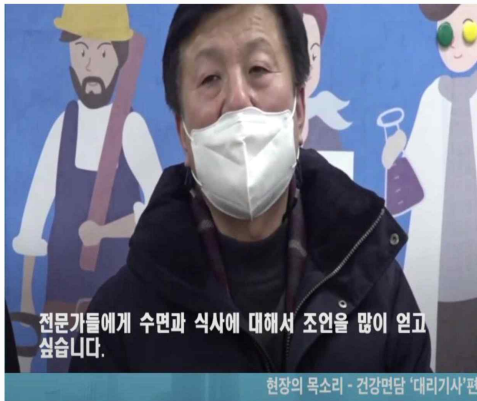
220413, 인천북구공단, 직종별건강진단비용지원, 대리기사2

On-site Health Examination + Offline contact → Open Examination

Occupational Health Examination by Safety and Health Agency
Implemented at shelters for mobile workers

Explain Health Examination Results

2



“

”

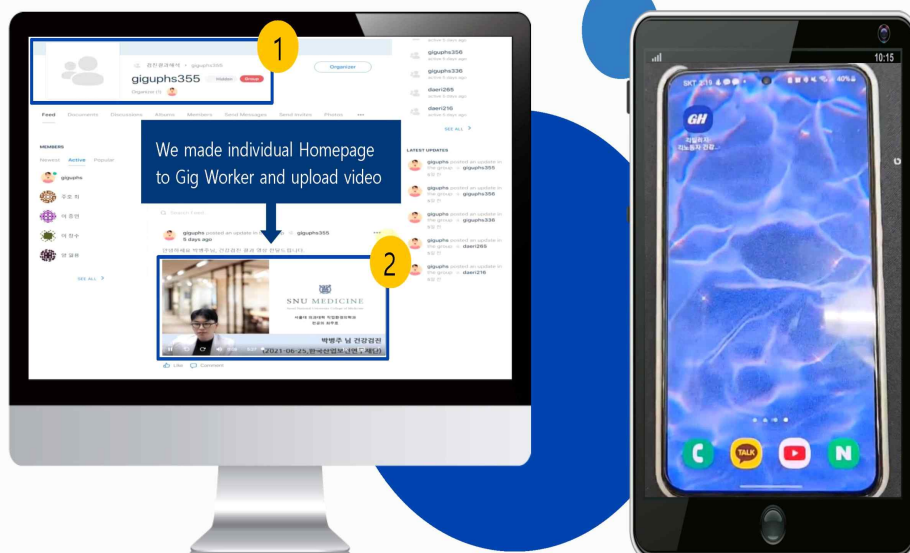
It is not easy to understand my health examination result paper, the hospital doctor not much time to explain to me.

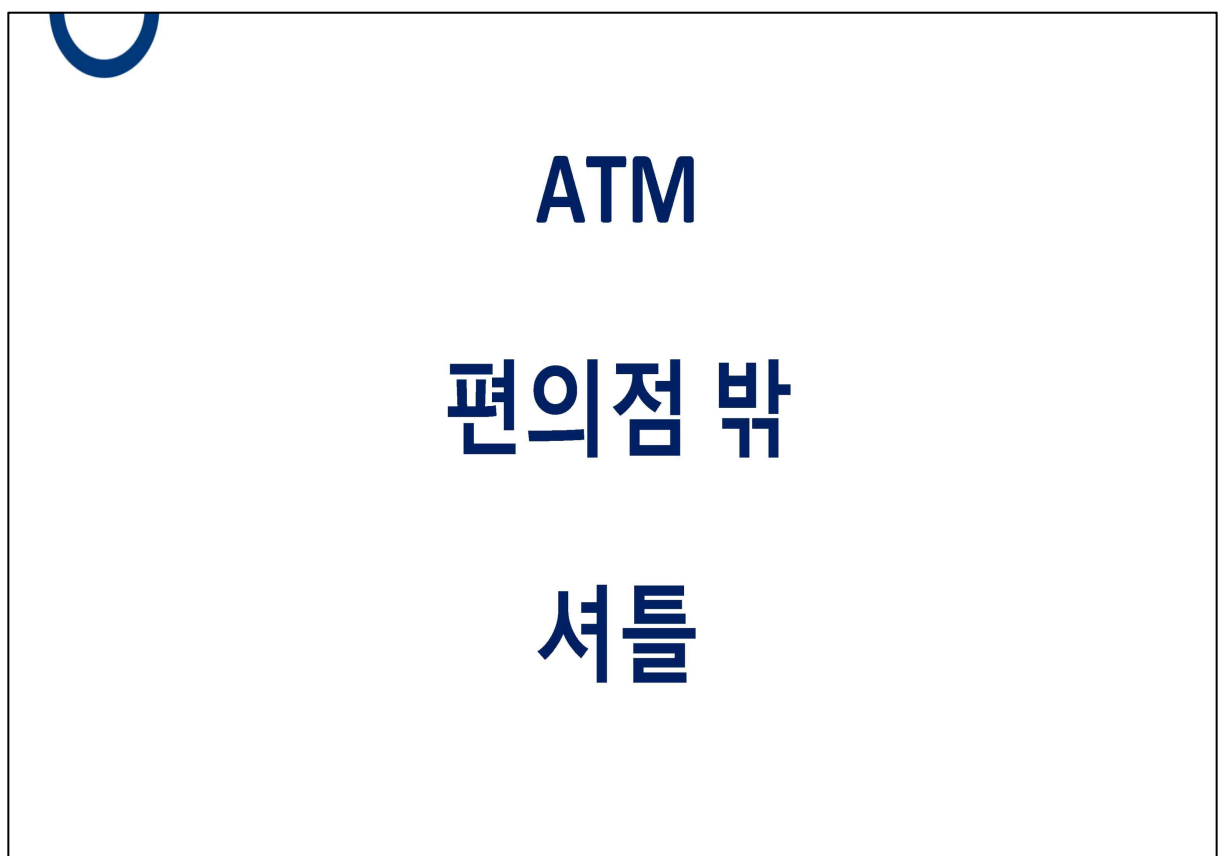
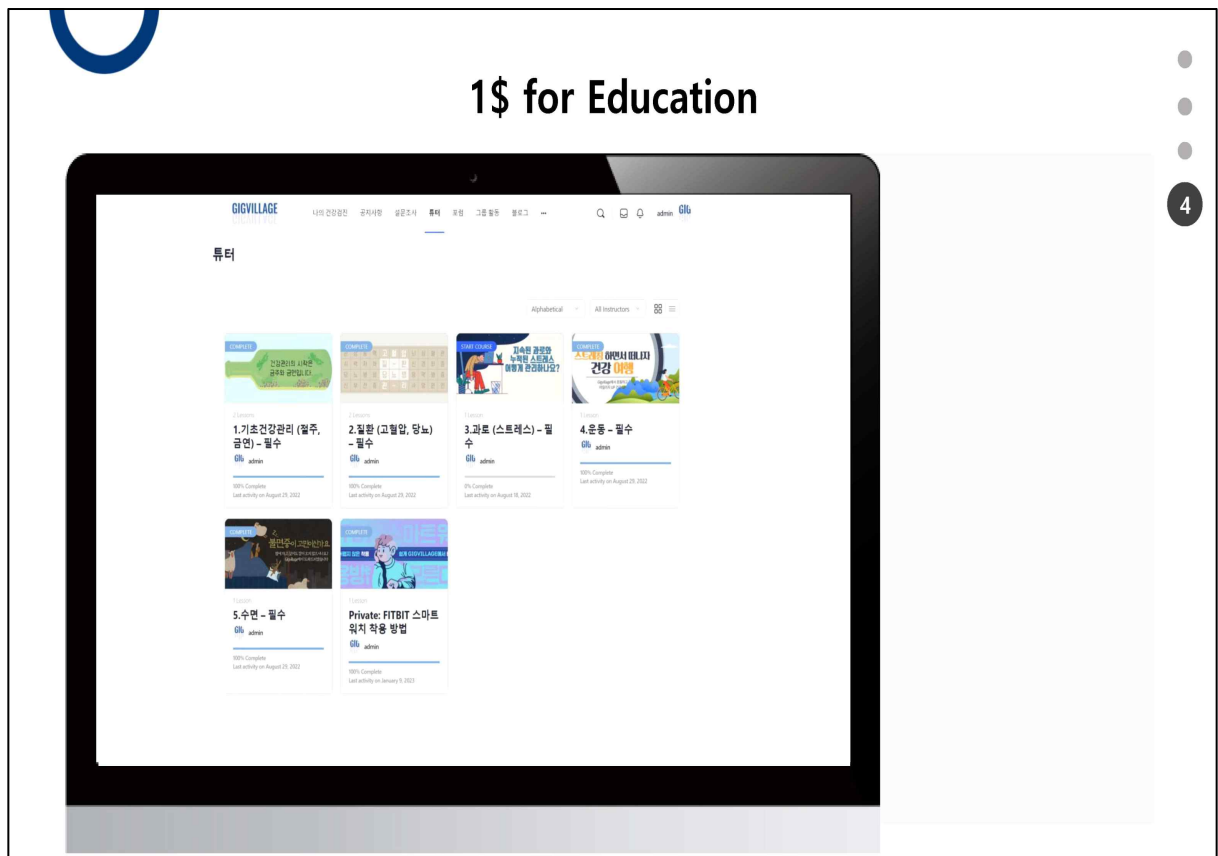
Should I go to hospital again? Will my health get worse? I don't know what I should do for my health.

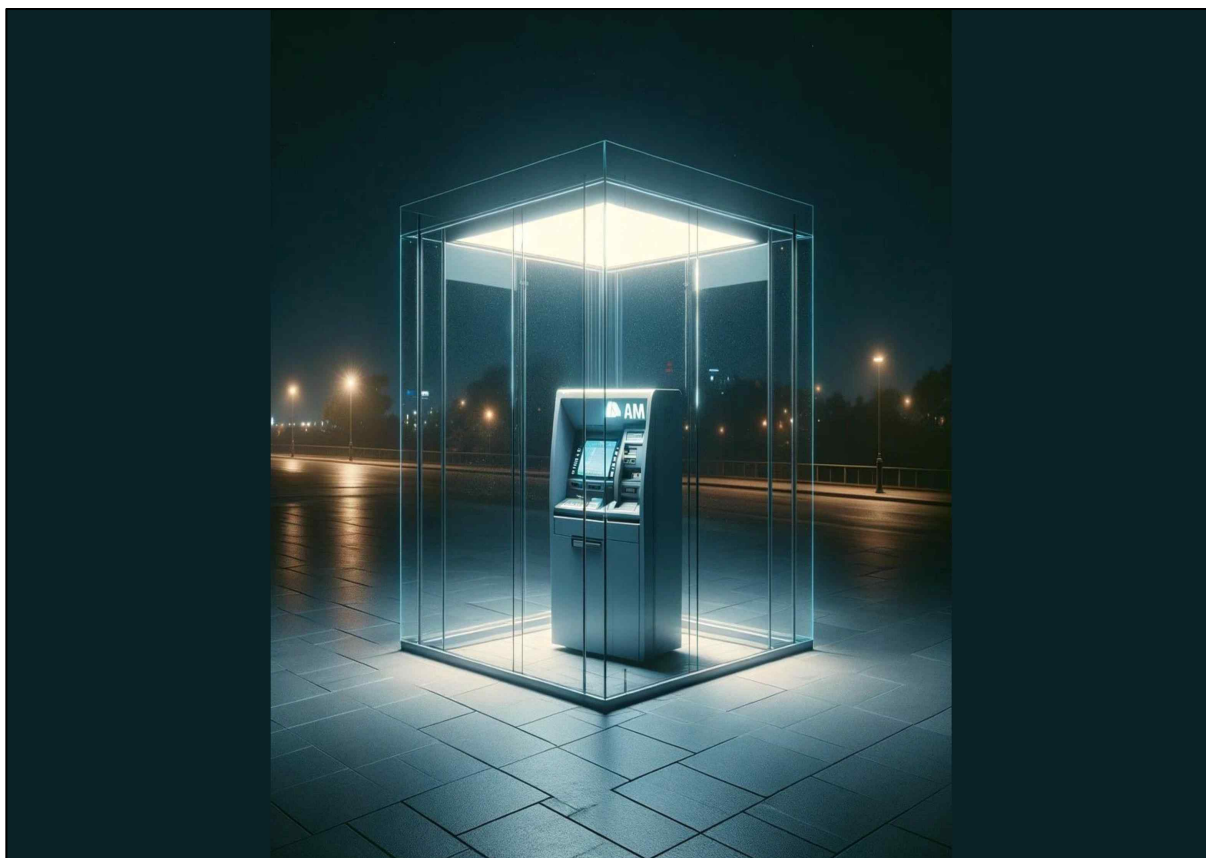
They suggested **repeatable video of doctor or nurse's message**. They want watch video again during working times.

Explain Health Examination Results

2









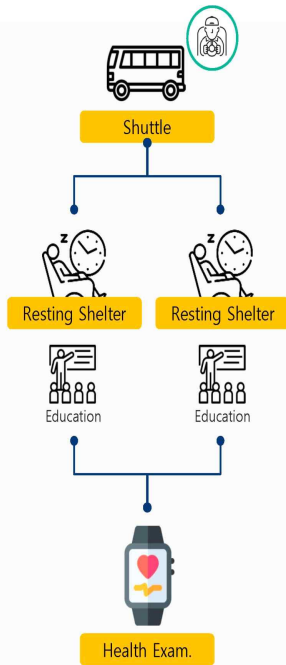
같은 공간, 같은 시간, 같은 생각



“ 3AM, there was a car accident. after car accident, ambulance took 4 passengers to hospital. The crashed car was towed to parking lot. In that morning, police men found a dead person under the very end seat. Because all passengers were Replacement driver, they did not know each other, and nobody knew that death.

There was no bus, no subway. I cannot walk to my home, because my last customer lived far away from my home. The taxi is too expensive for me to use. I usually used shuttle instead.

Night Shuttle

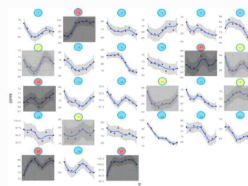


“ Replacement drivers fall into remote area late night where there is no public transportation running. So, we tried to make a night shuttle between remote area and Resting Shelter. The resting shelter can introduce mutual fund and health examination project for the shuttle users. ”

Night Shuttle



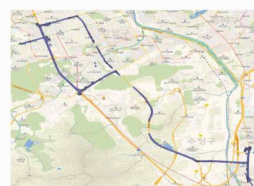
Num of Worker using Resting Shelters



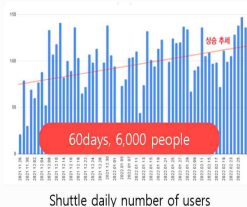
Heart rate variability in shuttle



Shuttle route(1)



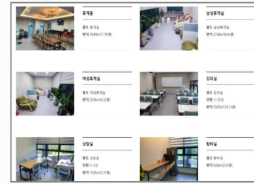
Shuttle route(2)



Shuttle daily number of users



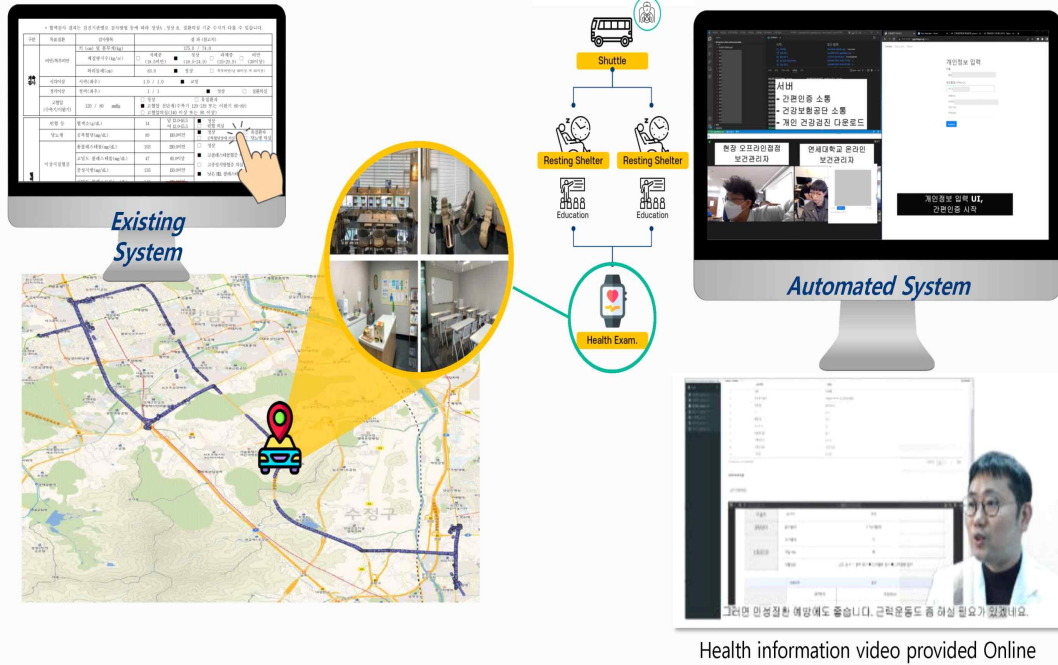
Shuttle in operation



“ **Government sustainable project**

The Platform cooperative team received a project for shuttle operation business expenses from the Ministry of Labor(300 million).

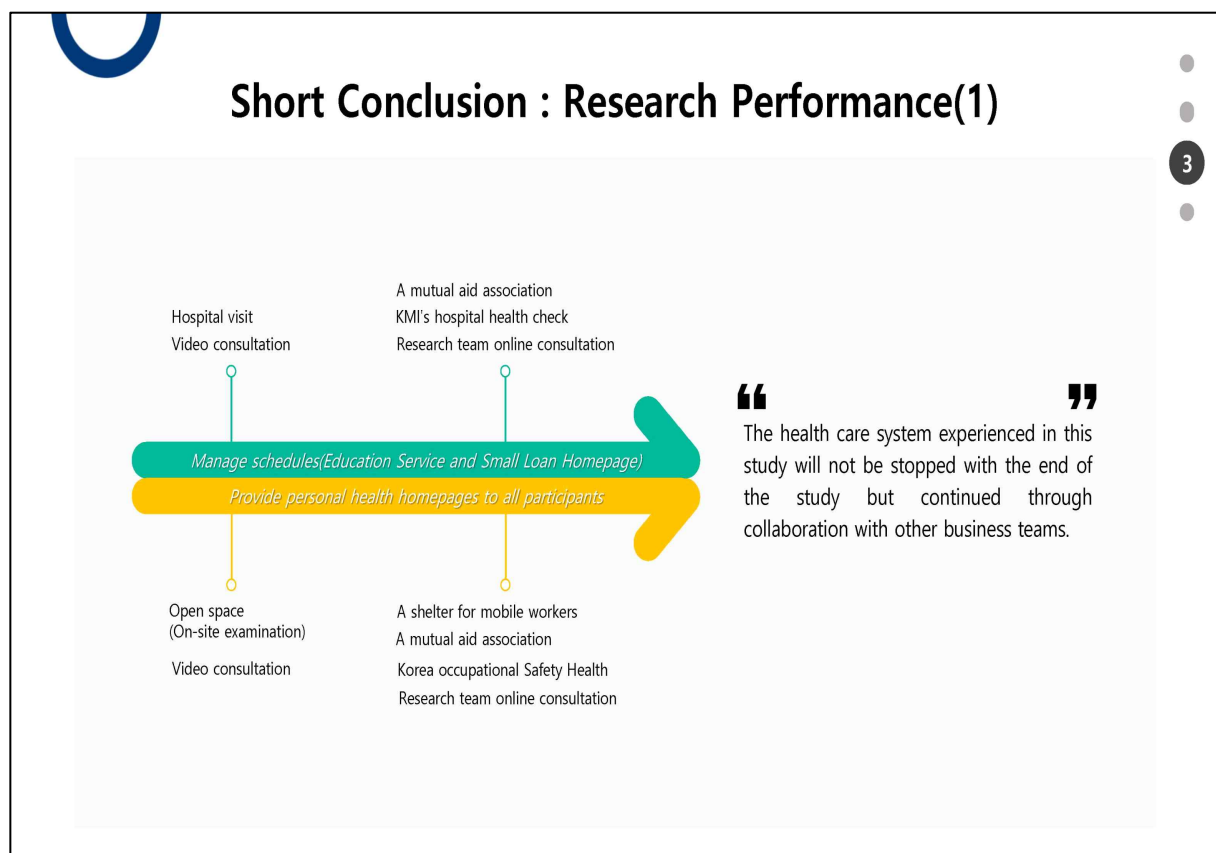
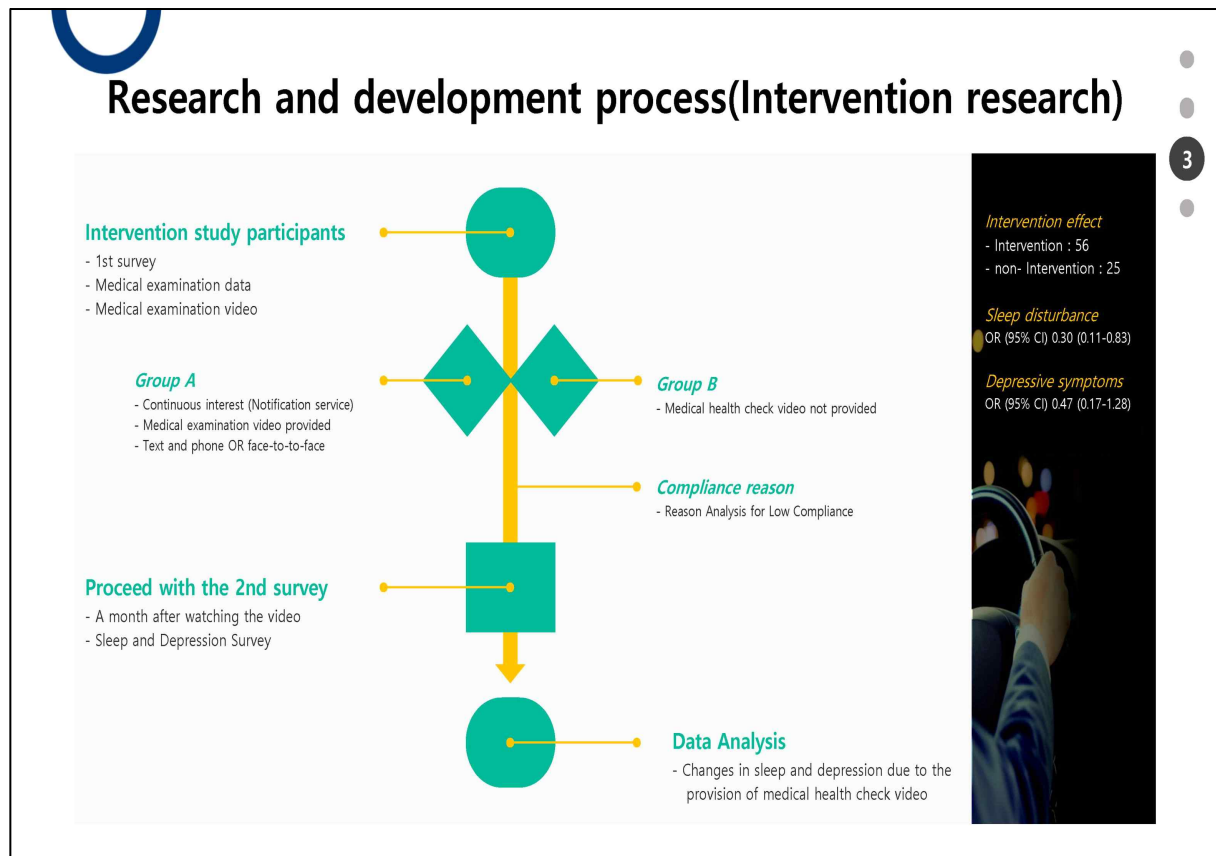
Night Shuttle(Provide Health information Online)



Automated System Development

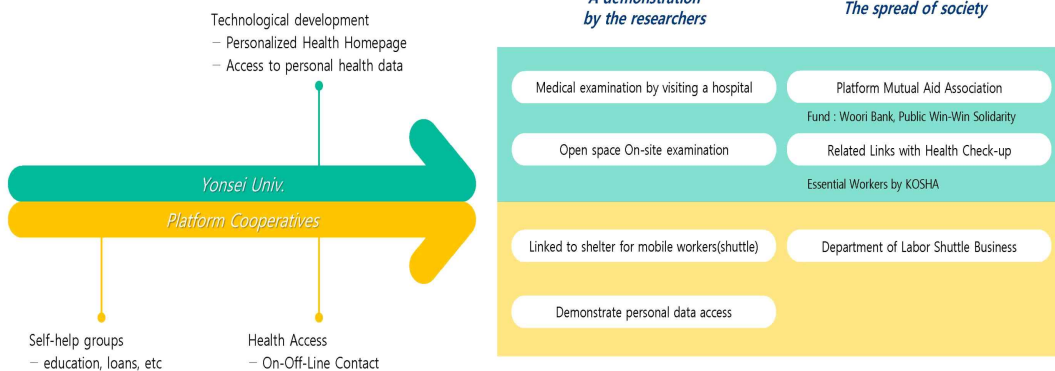
? Where are my Health Examination results?





Research and development comprehensive process

3



Others Data manage, Computer setting and programming

4

Developing Guide Book

Data management and analysis

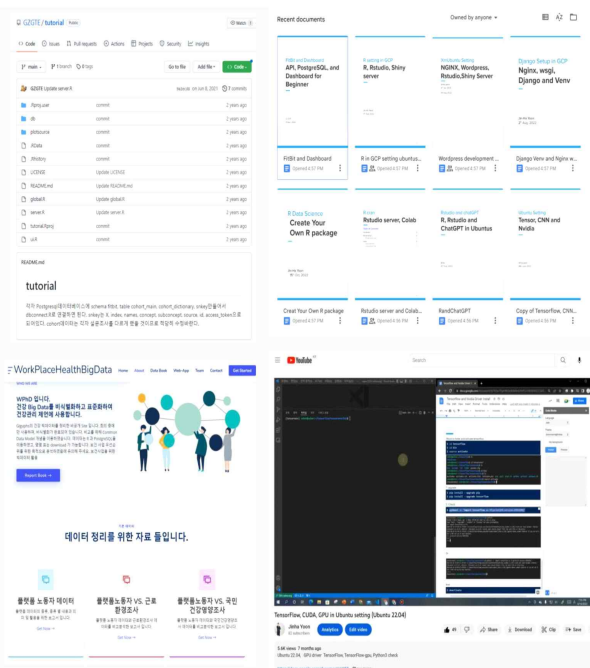
FitBit Heart Rate download, dashboard

NHIS-health examination

AutoReporting

Web and Mobile App

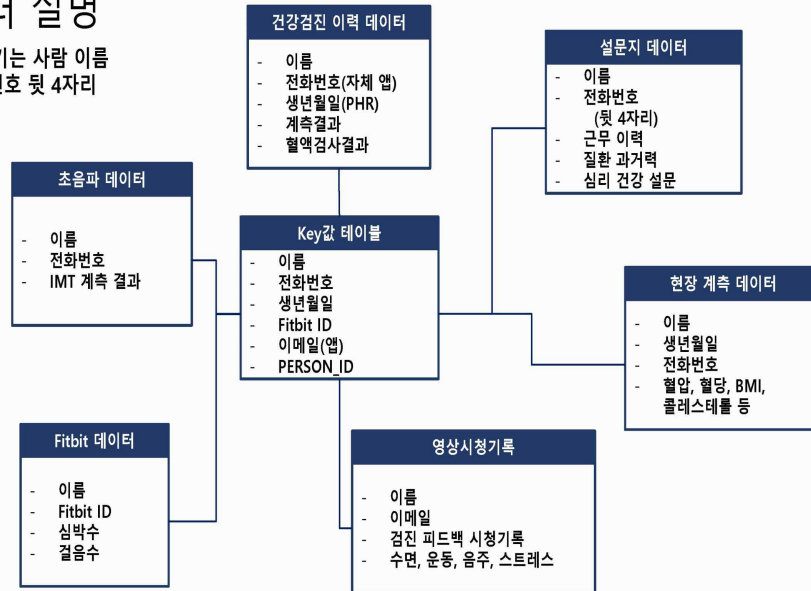
Prediction, analysis



Data Management and analysis

데이터 설명

기본 결합키는 사람 이름
+ 핸드폰번호 뒷 4자리



Data Management, and analysis

데이터 설명-결합예시

건강검진								
PERSON_ID	BIRTH_YEAR	SEX	FITBIT_ID	PERSON_ID	혈압	공복혈당	콜레스테롤	BMI
V1001	1961	1	123	V1001	135/78	112	205	25.5
V1002	1974	1	9999	V1003	154/98	139	178	26.3
V1003	1952	1	55					
V1004	1983	2	9999					
V1005	1991	1	9999					
V1006	1965	1	87					

설문									
PERSON_ID	우울 점수	불면증 점수	근무 이력 (시작)	근무 시간	PERSON_ID	우울 점수	불면증 점수	근무 이력 (시작)	근무 시간
V1001	12	14	2015년	주 50시간	V1001	12	14	2015년	주 50시간
V1005	5	10	2020년	주 68시간	V1005	5	10	2020년	주 68시간
V1006	21	18	2017년	주 40시간	V1006	21	18	2017년	주 40시간

FITBIT_ID	날짜	심박 수	걸음 수	FITBIT_ID	날짜	심박 수	걸음 수
123	2021.11.12	79	10988	123	2021.11.13	95	12154
55	2022.05.12	68	5900				

KEY TABLE

이름, 전화번호를 활용한
PERSON_ID 생성
E-mail 정보 호환

Data Management, and analysis

데이터 설명

설문조사

시범		1차/2차	
기본정보	9문항	기본정보	7문항
근무정보	7문항	근무정보	5문항
건강상태	52문항	건강상태	12문항
심리건강	45문항	심리건강	32문항
	- 수면 8문항		- 수면 8문항
	- 안과 7문항		- 삭제
	- 우울 9문항		- 우울 9문항
	- 폭행 12문항		- 폭행 6문항
사회적지지	13문항	사회적지지	6문항
	기타 9문항		기타 9문항
전체	126문항	전체	62문항

Data Management, and analysis

데이터 설명

데이터 코딩북 - 심리건강

수면	IS11	잠들기 어려움	0=없음, 1=약간, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제점이 얼마나 심한지 선택해 주십시오. (a. 잠들기 어려움)
	IS12	잠 유지 어려움	0=없음, 1=약간, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제점이 얼마나 심한지 선택해 주십시오. (b. 잠을 유지하기 어려움)
	IS13	너무 잠깐 잤다	0=매우 안쪽, 1=안쪽, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제점이 얼마나 심한지 선택해 주십시오. (c. 너무 잠깐 잤다)
	IS14	수면 만족도	0=매우 안쪽, 1=안쪽, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	3. 현재 당신의 수면패턴에 대해 얼마나 만족하고 있습니까?
	IS15	불안정 수면	0=전혀 그렇지 않음, 1=약간, 2=중간, 3=심하게, 4=매우 심하게	KWHE-CDM	4. 불면증이 일상생활을 어느 정도 방해한다고 생각합니까?
	IS16	주변 사람 생각	0=전혀 심각하게 생각하지 않음, 1=약간, 2=중간, 3=심하게, 4=매우 심하게	KWHE-CDM	5. 주변 사람들이 귀하의 수면문제에 대해 심각하게 생각하고 있습니까?
	IS17	불안정 수면	0=전혀 심각하게 생각하지 않음, 1=약간, 2=중간, 3=심하게, 4=매우 심하게	KWHE-CDM	6. 현재 불면증에 관하여 얼마나 걱정하고 있습니까?
안과	E_PEO	안과 수술 경험	1=예, 2=아니오, 99999=모름/무응답	KWCS	7. 과거에 안과 수술을 받은 적이 있습니까?
	E_DES_d9	안구건조 증상	1=예, 2=아니오, 99999=모름/무응답	KWCS	8. 안구건조증은 건조 또는 자극 증상 경험에 있습니까?
	E_DES_d9	안구건조 증상	1=예, 2=아니오, 99999=모름/무응답	KWCS	9. 안구건조증은 건조 또는 자극 증상을 받은 적이 있습니까?
	E_VS_D5	시력 장애	1=예, 2=아니오, 99999=모름/무응답	KWCS	10. 시력장애가 있습니까?
	E_VS_MYO	근시	1=예, 2=아니오, 99999=모름/무응답	KWCS	11. 멀리서 잘 보이지 않고, 가까운 것이 잘 보입니까?
	E_VS_MYPER	원시	1=예, 2=아니오, 99999=모름/무응답	KWCS	12. 멀리서 잘 보이고, 가까운 것이 잘 보입니까?
	E_VS_AST	복시	1=예, 2=아니오, 99999=모름/무응답	KWCS	13. 물체가 보일 때 두 개로 보입니까?
우울	BP_PHQ1	우울증 PHQ 설문 1	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	1. 기분을 밝게 하거나, 웃을 수 없습니까?
	BP_PHQ2	우울증 PHQ 설문 2	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	2. 평소 하던 일에 대한 흥미가 없어지거나 즐거움을 느끼지 못하십니까?
	BP_PHQ3	우울증 PHQ 설문 3	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	3. 잠들기가 어렵거나 자주 깬다고 느끼십니까?
	BP_PHQ4	우울증 PHQ 설문 4	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	4. 평소보다 식욕이 줄었습니까?
	BP_PHQ5	우울증 PHQ 설문 5	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	5. 다른 사람들이 눈치 채 정도보다 평소보다 많이 울었습니까? (혹은 너무 안절부절 못해서 기분이 답답할 수 있습니다)
	BP_PHQ6	우울증 PHQ 설문 6	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	6. 피곤하고 기운이 없습니다.
	BP_PHQ7	우울증 PHQ 설문 7	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	7. 내가 잘못했거나, 실수했다고 생각하십니까?
업무중 특정 피해	BP_PHQ8	우울증 PHQ 설문 8	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	8. 신문을 읽거나 TV를 보는 것과 같은 일상적인 일에도 집중할 수 없습니다.
	BP_PHQ9	우울증 PHQ 설문 9	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	9. 자라라 하는 것이 더 낫다고 생각하십니까?
	ai01_n	언어능력	연속형	KWCS (연속형)	1. 업무 수행 중에 언어능력이 몇 년 동안이었습니까?
	ai02_n	언어능력	연속형	KWCS (연속형)	2. 현재 언어 능력이 몇 년 동안이었습니까?
	ai03_n	언어능력	연속형	KWCS (연속형)	3. 현재 언어 능력이 몇 년 동안이었습니까?
	ai04_n	언어능력	연속형	KWCS (연속형)	4. 현재 언어 능력이 몇 년 동안이었습니까?
	ai05_n	언어능력	연속형	KWCS (연속형)	5. 현재 언어 능력이 몇 년 동안이었습니까?

4

4

Data Management, and analysis

데이터 설명

데이터 코딩북 - 심리건강

수면	ISI1	잠들기 어려움	0=없음, 1=약간, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제들이 얼마나 심한지 선택해 주십시오. (a. 잠들기 어려움)
	ISI2	잠 유지 어려움	0=없음, 1=약간, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제들이 얼마나 심한지 선택해 주십시오. (b. 잠을 유지하기 어려움)
	ISI3	너무 일찍 깨	0=없음, 1=약간, 2=중간, 3=심함, 4=매우 심함	KWHE-CDM	2. 최근 2주간 다른 각 항목의 문제들이 얼마나 심한지 선택해 주십시오. (c. 너무 일찍 깨는 문제)
	ISI4	수면 만족도	0=매우 만족, 1=만족, 2=중간, 3=불만족, 4=매우 불만족	KWHE-CDM	3. 현재 당신의 수면패턴에 대해 얼마나 만족하고 있습니까?
	ISI5	활상성향 방해	0=전혀 방해되지 않음, 1=약간, 2=중간, 3=심하게 방해, 4=매우 심하게 방해	KWHE-CDM	4. 불면증이 활상성향을 어느 정도 방해한다고 생각합니까?
	ISI6	주변 사람 생각	0=전혀 심각하게 생각하지 않음, 1=약간, 2=중간, 3=심각하게 생각, 4=매우 심각하게 생각	KWHE-CDM	5. 주변 사람들이 귀하의 수면문제에 대해 심각하게 생각하고 있습니까?
	ISI7	불면증 걱정	0=전혀 걱정하지 않는다, 1=약간, 2=중간, 3=심하게 걱정, 4=매우 심하게 걱정	KWHE-CDM	6. 현재 불면증에 관하여 얼마나 걱정하고 있습니까?
안과	E_PEO	안과 수술 경험	1=예, 2=아니오, 99999=모름/무응답	KWCS	7. 과거에 안과 수술을 받은 적이 있습니까?
	E_DES_dh	안구건조 증상	1=예, 2=아니오, 99999=모름/무응답	KWCS	8. 안구건조증(눈 건조 또는 자극) 증상 경험이 있습니까?
	E_DES_ldg	안구건조로 인한	1=예, 2=아니오, 99999=모름/무응답	KWCS	9. 안구건조증은 안과 또는 자극 진단을 받은 적이 있습니까?
	E_VS_DS	시력 장애	1=예, 2=아니오, 99999=모름/무응답	KWCS	10. 시력장애가 있습니까?
	E_VS_MHO	근시	1=예, 2=아니오, 99999=모름/무응답	KWCS	11. 멀리서 잘 보이고, 가까운 것이 잘 안보입니까?
우울	E_VS_MPER	원시	1=예, 2=아니오, 99999=모름/무응답	KWCS	12. 멀리서 잘 보이고, 가까운 것이 잘 안보입니까?
	E_VS_AST	복시	1=예, 2=아니오, 99999=모름/무응답	KWCS	13. 물건이 흐리게 보이거나 겹쳐보입니까?
	BP_PHQ1	우울증 PHQ 설문 1	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	1. 기분이 가라앉거나, 우울하거나, 희망이 없다고 느꼈다.
	BP_PHQ2	우울증 PHQ 설문 2	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	2. 평소 하던 일에 대한 흥미가 없어지거나 즐거움을 느끼지 못했다.
	BP_PHQ3	우울증 PHQ 설문 3	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	3. 잠들기가 어렵거나 자주 깨다. 혹은 너무 많이 잤다.
	BP_PHQ4	우울증 PHQ 설문 4	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	4. 평소보다 식욕이 줄었다. 혹은 너무 많이 먹었다.
	BP_PHQ5	우울증 PHQ 설문 5	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	5. 다른 사람들이 눈치 잘 찜토록 평소보다 활동이 느려졌다. 혹은 너무 활동부진 못해서 기분이 안 좋을 수 있었다.
	BP_PHQ6	우울증 PHQ 설문 6	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	6. 과민하고 기분이 안 좋았다.
	BP_PHQ7	우울증 PHQ 설문 7	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	7. 내가 잘못했거나, 실수했다는 생각이 들었다. 혹은 자신과 가족을 실망시켰다고 생각했다.
	BP_PHQ8	우울증 PHQ 설문 8	0=없음, 1=2-6월, 2=7-12월, 3=3개월, 99999=무응답	KNHANES	8. 신문을 읽거나 TV를 보는 것과 같은 일상적인 일에도 집중할 수가 없었다.
임종증 특성 이해	aiab1_n	인간관계	연속형	KWCS (전수조사)	1. 업무 상황 중에 인간관계가 잘 안 됐습니까?
	aiab2_n	원하지 않는 성격 관심	연속형	KWCS (전수조사)	2. 원하지 않는 성격 관심이 몇 건 있었습니까?
	aiab3_n	위협	연속형	KWCS (전수조사)	3. 위협 또는 공격적 행동이 몇 건 있었습니까?
	aiab4_n	인격모독	연속형	KWCS (전수조사)	4. 인격모독을 몇 건 있었습니까?
	aiab5_n	신체적 폭력	연속형	KWCS (전수조사)	5. 신체적 폭력이 몇 건 있었습니까?
	aiab6_n	성폭행	연속형	KWCS (전수조사)	6. 업무 외 일상생활, 사귀기 등의 상황이 몇 건 있었습니까?
	aiab7_yh	인간관계 관련 상태	핵심변수, 2= 고위, 3=대형업체(또는 사무실), 4= 프리랜서 사	KWCS	7. 위 사건은 누구에게 있었습니까? (중복가능) (1. 언어폭력)

Data Management, and analysis

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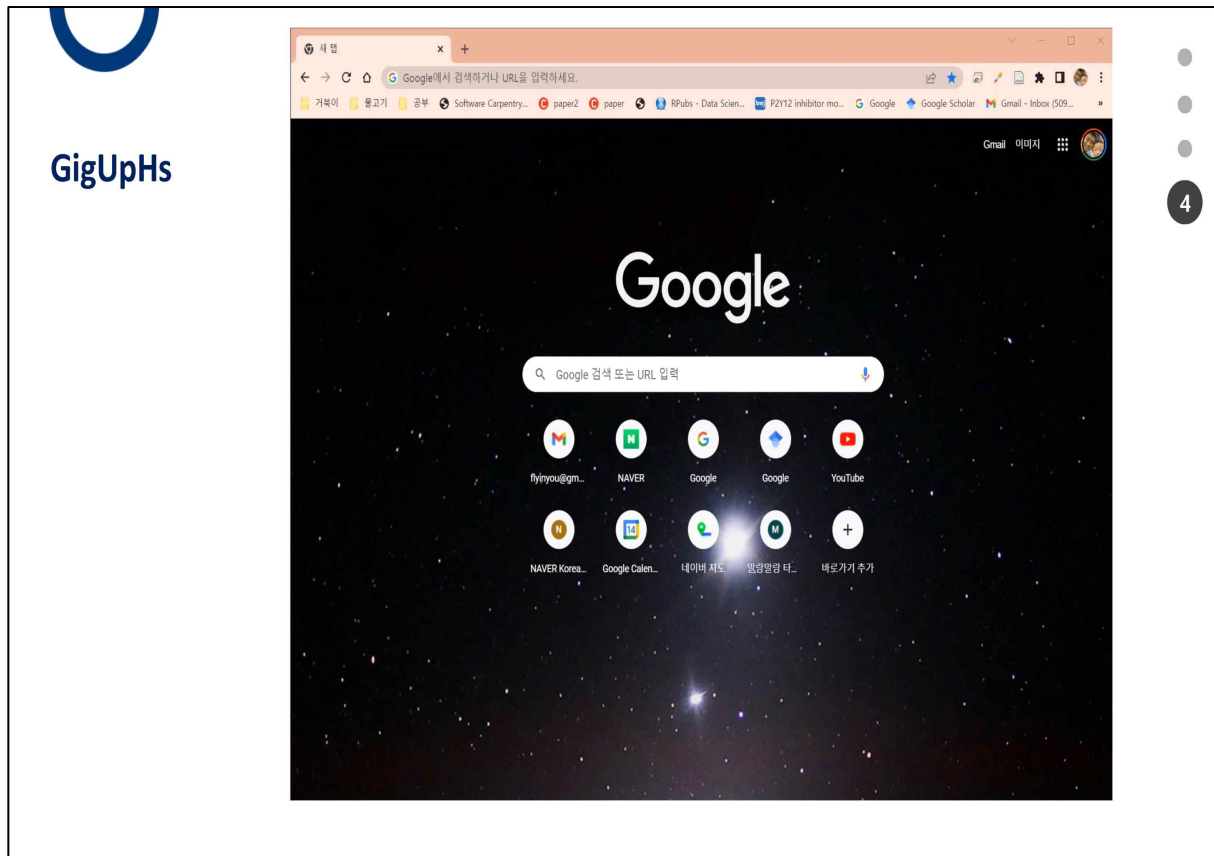
데이터 분석 **Baseline Characteristics** Logistic Model Plot

variables	values	0.non	distribution	sleep	distribution	prop	p-value
1 age	45.812.4	45.710.9	<0.001				
2 sex	Men	17900 (95.3%)	1527 (6.6%)	<0.001			
3 sex	Women	19848 (94.7%)	2041 (9.3%)				
4 wttime_week		42.512.8	43.014.5	0.042			
5 whaltime	balance	21075 (94.8%)	2224 (9.3%)	<0.001			
6	non balance	726 (8.2%)	1014 (12.3%)				
7 edup	university or more	19297 (94.4%)	1502 (7.1%)	<0.001			
8	high school	14029 (94.4%)	1248 (8.7%)				
9	middle school or below	2400 (94.4%)	548 (14.6%)				
10 empp	paid worker	25784 (94.4%)	2122 (7.6%)	<0.001			
11	employee/self-employed	25238 (94.4%)	2249 (8.7%)				
12	own account worker	8327 (94.4%)	880 (9.3%)				
13	unpaid family work	1039 (94.4%)	127 (12.3%)				
14 vovtsgp	Never	12221 (94.4%)	684 (5.5%)	<0.001			
15	Rarely	12228 (94.4%)	714 (5.5%)				
16	Sometimes	9101 (94.4%)	981 (9.3%)				
17	Often	3423 (94.4%)	775 (12.3%)				
18	Always	614 (7.0%)	272 (10.0%)				
19 skillWork	non skill work	23024 (94.4%)	3073 (8.4%)	<0.001			
20	skill work	20784 (94.4%)	2195 (9.4%)				
21	non job	27407 (94.4%)	3371 (12.3%)	<0.001			
22	job	2894 (8.4%)	30 (15.6%)				
23	whgp	135	2642 (94.4%)	269 (9.3%)			
24	140	1124 (94.4%)	131 (10.4%)				
25	135	1424 (94.4%)	1912 (12.3%)				
26	140	1274 (94.4%)	219 (9.3%)				
27	140	4221 (94.4%)	516 (10.1%)				

Gigapops



2. 디지털 기술을 활용한 플랫폼 노동자의 스마트 건강관리



Data Management, and analysis

교육 프로그램 기록 데이터

연월일	학년	이수 여부	N (%)
55세 이하	고등학교 이하	아니오	19 (55.88%)
55세 이하	고등학교 이하	예	15 (44.12%)
55세 이하	대학교 이상	아니오	26 (42.62%)
55세 이하	대학교 이상	예	35 (57.38%)
55세 초과	고등학교 이하	아니오	12 (73%)
55세 초과	고등학교 이하	예	4 (25%)
55세 초과	대학교 이상	아니오	15 (35.71%)
55세 초과	대학교 이상	예	27 (64.29%)

교육 ID	E-mail	이수 교육 과정명	교육 이수 여부	교육 생성 시간
148	daw105@giguphs.org	2106_건강관리교육	1	2022-12-01
150	daw107@giguphs.org	2031_건강관리교육	1	2022-12-01
205	daw119@giguphs.org	2074_건강관리교육	1	2022-12-02
210	daw120@giguphs.org	2749_건강관리교육	1	2022-11-25
247	daw130@giguphs.org	2609_건강관리교육	1	2022-11-25
246	daw133@giguphs.org	2764_건강관리교육	1	2022-11-22
271	daw176@giguphs.org	2164_건강관리교육	1	2022-11-28
276	daw180@giguphs.org	2179_건강관리교육	1	2022-11-28
311	daw190@giguphs.org	2618_건강관리교육	1	2022-12-02
498	daw192@giguphs.org	2822_건강관리교육	1	2022-11-17
239	daw193@giguphs.org	2132_건강관리교육	1	2022-12-08
401	daw194@giguphs.org	100_7차건강관리(영양·운동·흡수·활동)	2	2022-08-29
401	daw195@giguphs.org	99_2차건강관리(영양·운동·흡수·활동)	2	2022-08-29
401	daw196@giguphs.org	99_3차건강관리(영양·운동·흡수·활동)	1	2022-08-29
401	daw197@giguphs.org	94_4차건강관리(영양·운동·흡수·활동)	1	2022-08-29
401	daw198@giguphs.org	92_5차건강관리(영양·운동·흡수·활동)	1	2022-08-29
401	daw199@giguphs.org	134_7차건강관리(영양·운동·흡수·활동)	1	2022-08-29
401	daw200@giguphs.org	2623_건강관리교육	1	2022-11-02
478	daw201@giguphs.org	2626_건강관리교육	1	2022-12-02
249	daw202@giguphs.org	2017_건강관리교육	1	2022-11-18
483	daw203@giguphs.org	2504_건강관리교육	1	2022-11-05
402	daw204@giguphs.org	99_2차건강관리(영양·운동·흡수·활동)	2	2022-08-31
402	daw205@giguphs.org	2514_건강관리교육	1	2022-12-06
474	daw206@giguphs.org	2524_건강관리교육	1	2022-11-25
103	daw207@giguphs.org	2087_건강관리교육	1	2022-11-28
252	daw208@giguphs.org	2027_건강관리교육	1	2022-11-28
501	daw209@giguphs.org	2891_건강관리교육	1	2022-12-07
497	daw210@giguphs.org	2900_건강관리교육	1	2022-11-24
454	daw211@giguphs.org	2531_건강관리교육	1	2022-11-16
456	daw212@giguphs.org	2541_건강관리교육	1	2022-12-02
232	daw213@giguphs.org	2546_건강관리교육	1	2022-12-07
132	daw214@giguphs.org	2496_건강관리교육	1	2022-12-02
496	daw215@giguphs.org	2896_건강관리교육	1	2022-11-26

Data Management, and analysis

The Association between Replacement Drivers and Depressive Symptoms

Jongmin Lee¹, Heejoo Park², Juyeon Oh³, Juho Sim³, Choram Lee⁴, Yangwook Kim³, Byungyeon Yun^{4,5,*} and Jin-Ha Yoon^{4,5,*}

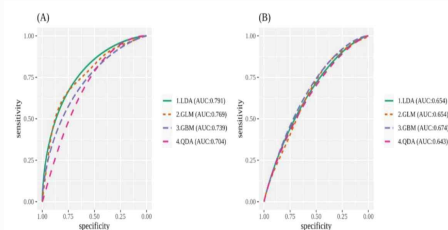


Figure 3. Differences in the area under the curve (AUC) based on whether (A) replacement driver variable was used or (B) not.

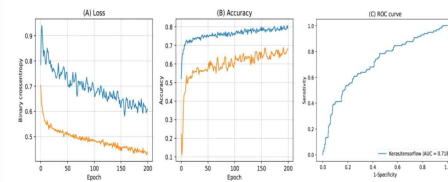


Figure 4. (A) Loss, (B) Accuracy, and (C) ROC curve of the deep learning model with Tensorflow.

This research was funded by a grant from the Korea Health Promotion Institute R&D Project funded by the Ministry of Health & Welfare, Republic of Korea (grant number: HS21C2387)

Variables	Model 1	Model 2	Final model
Replacement driver			
No	Reference (1.00)	Reference (1.00)	Reference (1.00)
Yes	9.09 (6.49–12.74)	8.59 (6.10–12.09)	7.89 (5.53–11.26)
Age	0.98 (0.96–0.99)	0.98 (0.96–0.99)	0.97 (0.96–0.99)
Education			
Below high school		Reference (1.00)	Reference (1.00)
Above university		0.94 (0.68–1.29)	0.83 (0.60–1.15)
Household income			
High		Reference (1.00)	Reference (1.00)
Low		1.34 (1.00–1.80)	1.31 (0.97–1.77)
Smoking			
None			Reference (1.00)
Ex-smoker			1.34 (0.88–2.05)
Current smoker			1.84 (1.22–2.78)
Alcohol consumption			
None			Reference (1.00)
Social drink			1.25 (0.80–1.96)
Heavy			1.36 (0.85–2.18)
Working hours			
Short (<40)			Reference (1.00)
Long (≥40)			1.18 (0.88–1.57)
Muscular exercise			
None			Reference (1.00)
Exerciser (>1 day)			1.60 (1.16–2.21)
Sleeping time			
Normal (6–9)			Reference (1.00)
Abnormal (≤5 / ≥10)			1.86 (1.28–2.71)

Data Management, and analysis

Common Data Model Presenteeism/Absenteeism

근로환경조사의 일반 임금근로자와 비교

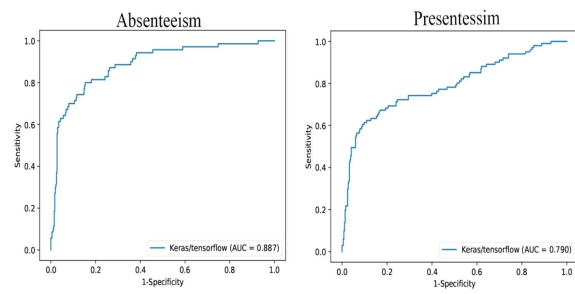
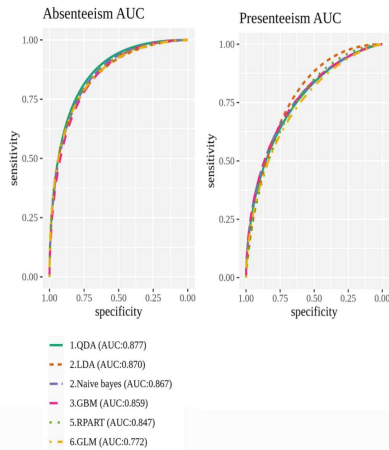
Table 2. Absenteeism & Presenteeism

Variable	Values	Presenteeism Model			Absenteeism Model		
		Odds Ratios	CI	p	Odds Ratios	CI	p
Platform	the general population	1.00 (reference)			1.00 (reference)		
	Platform worker	6.70	4.26–10.51	<0.001	26.91	15.33–47.22	<0.001
Age		1.00 (reference)			1.00 (reference)		
	High	1.00	0.99–1.01	0.88	1.00	0.98–1.02	0.98
Education	High	1.14	0.86–1.51	0.38	0.96	0.68–1.35	0.80
	Low	1.00 (reference)			1.00 (reference)		
	High	0.80	0.55–1.15	0.23	0.94	0.59–1.50	0.81
Income	Medium	0.52	0.34–0.79	0.002	0.81	0.49–1.33	0.40
	Low	1.00 (reference)			1.00 (reference)		
Working hour	< 40 hour	1.68	1.16–2.42	0.006	1.41	0.92–2.16	0.11
	≥ 40 hour	1.00 (reference)			1.00 (reference)		
Verbal Abuse	No	1.00 (reference)			1.00 (reference)		
	Yes	0.83	0.50–1.36	0.45	0.64	0.36–1.12	0.12
Sexual Violence	No	1.00 (reference)			1.00 (reference)		
	Yes	2.76	1.17–6.51	0.02	1.75	0.77–3.96	0.18
Physical Violence	No	1.00 (reference)			1.00 (reference)		
	Yes	3.29	1.90–5.71	<0.001	3.66	2.03–6.53	<0.001
Working days for week	> 5day	1.12	0.78–1.62	0.54	1.08	0.70–1.66	0.73
	≤ 5day	1.00 (reference)			1.00 (reference)		
working duration	High	1.36	0.96–1.91	0.08	1.62	1.06–2.46	0.03
	Medium	1.70	1.16–2.49	0.006	1.57	0.97–2.52	0.06
	Low	1.00 (reference)			1.00 (reference)		
Hard sleep	No	0.63	0.37–1.07	0.09	1.14	0.59–2.21	0.7
	Yes	1.00 (reference)			1.00 (reference)		
Maintain sleep	No	1.47	0.91–2.39	0.11	0.52	0.26–1.02	0.06
	Yes	1.00 (reference)			1.00 (reference)		
Absenteeism	Not difficult						
	Difficult						

Adjusted by age, edu, income, working hour, verbal abuse, sexual violence, physical violence, day5, working duration, hard sleep and maintain sleep.

Data Management, and analysis

데이터 분석 결과-Presenteeism/Absenteeism



- 전체적으로 Absenteeism과 Presenteeism의 예측 모델의 AUC값이 높게 나타남
- Absenteeism: QDA의 AUC 0.877, Presenteeism: GBM의 AUC 0.798

Data Management, and analysis

Unmet Health Care - Hypertension

연령대	고혈압	대조군	플랫폼
55세 이하	아니오	609(76.22%)	39(65%)
55세 이하	예	190(23.78%)	21(35%)
55세 초과	아니오	168(49.56%)	24(40.68%)
55세 초과	예	171(50.44%)	35(59.32%)

고혈압
Age-adjusted OR 1.26
(0.85-1.87; p=0.25)

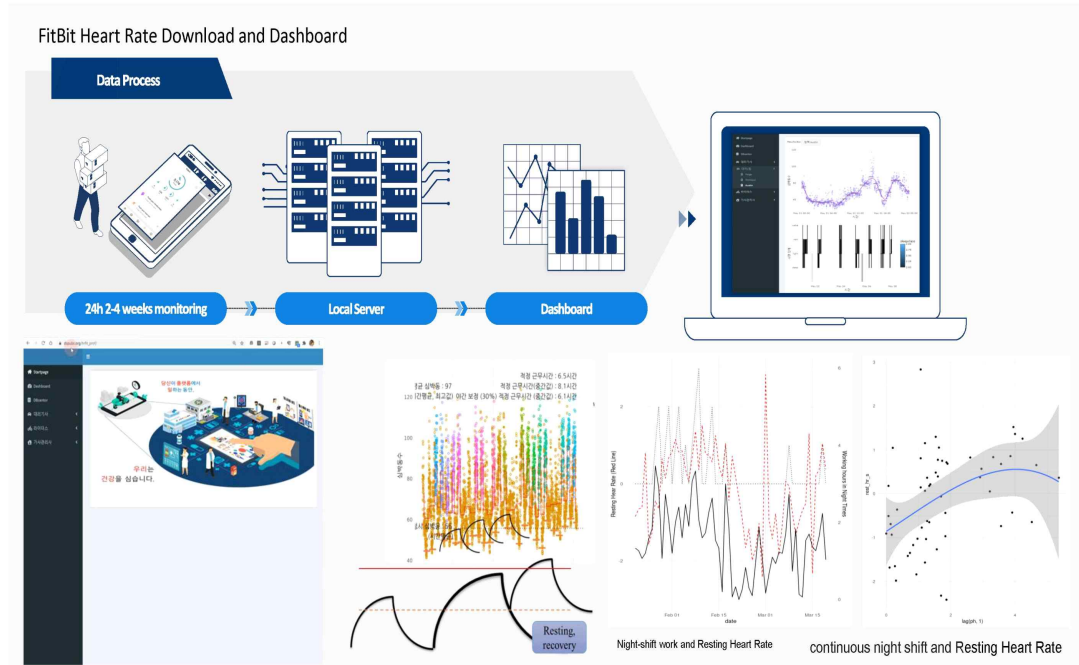
고혈압 중 조절/관리 X
Age-adjusted OR 8.43
(3.48-20.39; p<0.001)

연령대	고혈압 과거력	측정시 고혈압	대조군	플랫폼
55세 이하	아니오	예	107(56.32%)	17(80.95%)
55세 이하	예	아니오	54(28.42%)	1(4.76%)
55세 이하	예	예	29(15.26%)	3(14.29%)
55세 초과	아니오	예	30(17.54%)	25(71.43%)
55세 초과	예	아니오	107(62.57%)	5(14.29%)
55세 초과	예	예	34(19.88%)	5(14.29%)

➡ 고혈압 인지 X

➡ 고혈압 조절 X

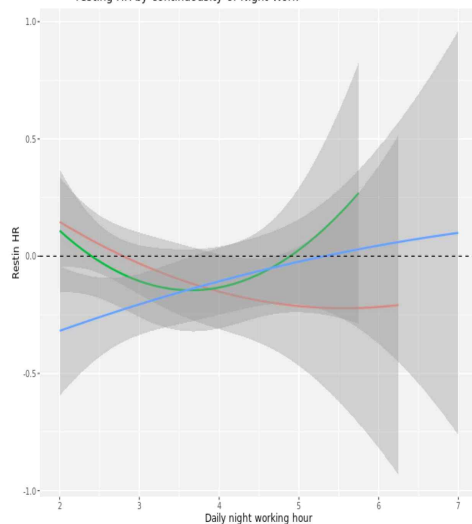
Data Management, and analysis



(플랫폼 노동자 건강증진 연구개발 시연영상자료)

Data Management, and analysis

Association between daily night working hour and resting HR by continuity of Night Work



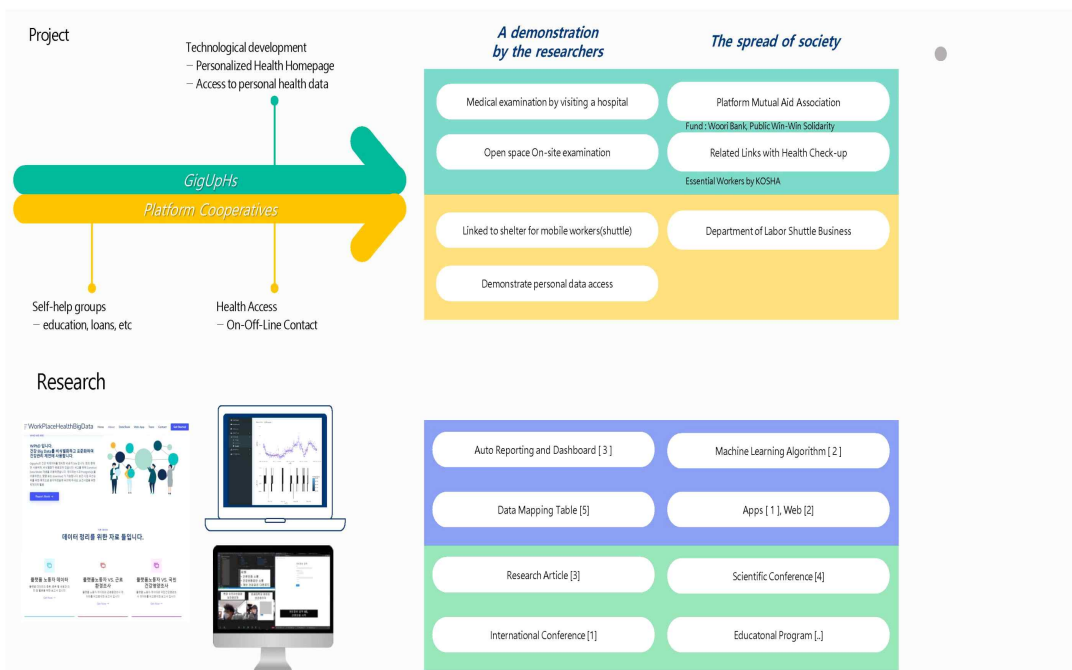
Discontinuous
Continuous(2days)
Continuous(≥3days)

Table 4. Odds ratios and 95% confidence interval of resting HR elevation by daily night working hour and continuity of night work

Daily night working hour	Discontinuous night work	Continuous night work		
		2 days	3 days or more	Overall
2 ≤ <3	Ref	Ref	Ref	Ref
3 ≤ <4	0.95(0.54-1.66)	1.12(0.49-2.55)	1.26(0.66-2.41)	1.16(0.70-1.91)
4 ≤ <5	0.65(0.32-1.30)	0.75(0.29-1.93)	1.48(0.79-2.8)	1.16(0.70-1.94)
5 ≤	0.68(0.23-2.01)	2.21(0.57-8.64)	2.76(1.19-6.36)	2.42(1.21-4.87)

*Adjusted for age

Summary



Thank You

GigUpHs

본 연구는 국민건강 스마트 관리 연구개발사업의 일환으로 이루어졌습니다.

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같은 공간, 같은 시간, 같은 생각

